Chapter 2 exercises

Work over $\mathbb{R}$ unless specified otherwise. Show row operations or their multipliers.

CH02-E1

Solve

$$\begin{aligned} x &+ 3y &&= 7, \\ 2x &- y &&= 0. \end{aligned}$$

Interpret the answer using the row and column pictures.

CH02-E2

For

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 4 \\ 0 & 4 & 3 \end{bmatrix},$$

perform Gaussian elimination. List the three pivots and the multipliers, and decide whether A is invertible.

CH02-E3

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

1. State the shape of AB.
2. Compute AB.
3. Determine whether BA exists. If it does, compute it and compare its shape with AB.

CH02-E4

Find the inverse of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

using row operations on $[A \mid I]$. Verify $A^{-1}A = I_2$.

CH02-E5

For

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$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix},$$

find $A = LU$ with unit lower triangular L, and verify the product.

CH02-E6

Let $x = (1, 2)^T$ and $y = (3, -1)^T$.

1. Compute $x^T y$ and state its type.
2. Compute xy^T and state its type.
3. Find the rank of xy^T and explain it through the column span.

CH02-E7

Extend elimination to function spaces. On $C([-1, 1], \mathbb{R})$, use

$$\langle f, g \rangle = \int_{-1}^{1} f(t)g(t)\, dt.$$

1. For $p_1 = 1$, $p_2 = 1 + t$, compute the Gram matrix G, prove positive definiteness, and find $\det G$.
2. Express $f(t) = 1 + 3t$ as $c_1 p_1 + c_2 p_2$. State and solve the Gram system, then substitute back.
3. On $[0, 2\pi]$, project $f(x) = 1 + x$ onto $\operatorname{span}\{1, \cos x, \sin x\}$. Compute each coefficient and the finite Fourier projection f_1.
4. Verify that $e = f - f_1$ is orthogonal to all three basis functions.

CH02-E8

For $a \in \mathbb{R}$ and a positive integer n, let $T_n(a)$ be the symmetric $n \times n$ Toeplitz matrix with diagonal a, adjacent diagonals 1, and all other entries 0.

1. Let $D_n = \det T_n(a)$, $D_0 = 1$. Expand along the first row to prove $D_n = aD_{n-1} - D_{n-2}$ for $n \geq 2$.
2. Compute D_1, D_2, D_3 and characterize invertibility of $T_3(a)$.
3. Let $\theta_k = k\pi/(n+1)$, and define component j of eigenvector k as $\sin(j\theta_k)$, for $j = 1, \ldots, n$. Prove $T_n(a)v_k = (a + 2\cos\theta_k)v_k$.
4. Deduce an eigenvalue product for $D_n(a)$ and check every factor for $n = 3, a = 3$.

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CH02-E9

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$.

1. Compute AB, BA, their dimensions, and their traces.
2. Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for compatible real matrices from the entry formula.
3. Explain why real square matrices cannot satisfy $AB - BA = I_n$. Does the same argument work for $n = 2$ over a field of characteristic 2?

CH02-E10

Let

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 3 \\ 6 & 4 & 5 \end{bmatrix}, \qquad b = (3, 8, 12)^T.$$

1. Use $a_{11} = 2$ as the first pivot and eliminate below it. Write the resulting matrix and explain why the second step needs a row swap.
2. Swap rows 2 and 3. Find P, L, U and verify $PA = LU$. Explain how the stored first-column multipliers move with the swap.
3. Solve $Ly = Pb$, $Ux = y$, and substitute into the original equations.
4. Compute $\det A_1, \det A_2, \det A$. Explain why invertibility of A does not guarantee LU without swaps.

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References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 2.

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